Intelligence Academy

Scikit-Learn 101

03. Estimators, Transformers, Predictors

Mejbah Ahammad

Module Agenda: The Shared Interface

We will break down the Scikit-Learn "Trinity"—the three objects that power the entire library.

1. The Estimator — `The Parent (fit)`

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2. The Transformer — `Modifies Data (X → X')`

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3. The Predictor — `Generates Labels (X → y)`

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4. Optimization — `fit_transform() Shortcut`

1. The Estimator

1.1 The Blueprint: Anatomy of an Estimator

Every algorithm inherits from `BaseEstimator`. This provides the "DNA" for the model.

sklearn.base.BaseEstimator

```
Defines:
+ get_params()
+ set_params()
```

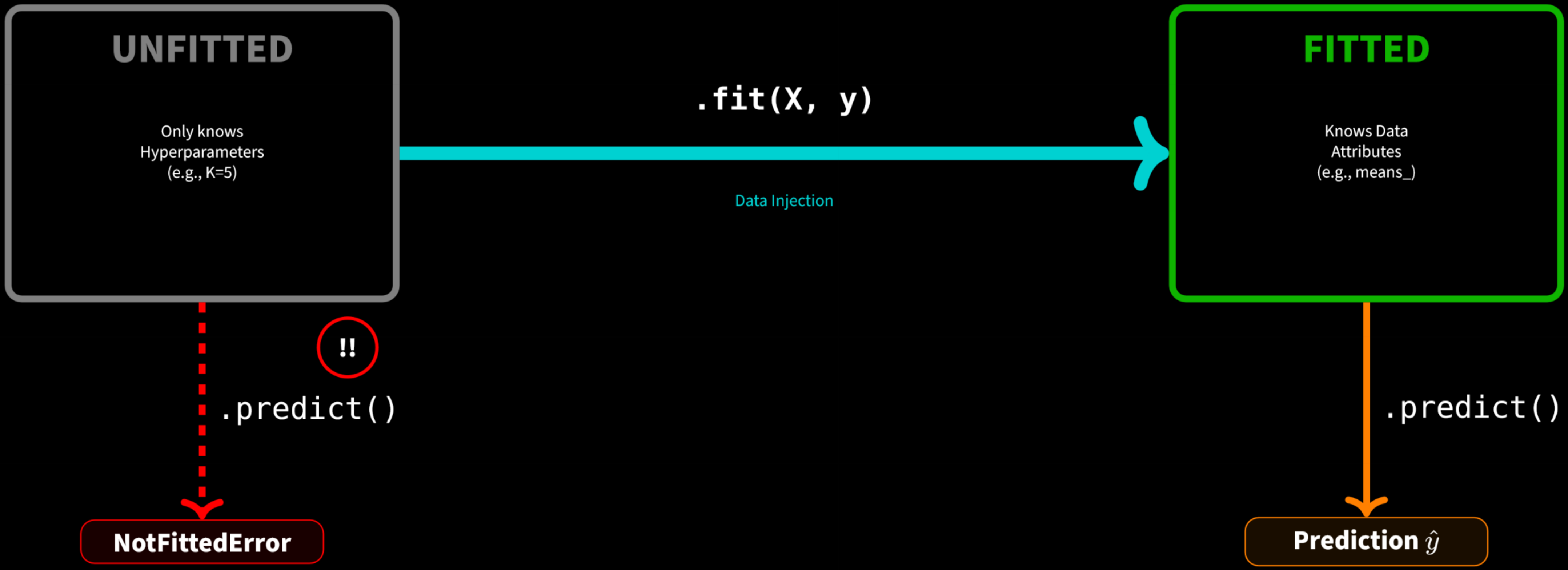
Inherits Methods

YourModel (The Child)

```
def __init__(self, K=5):
 # 1. Store Config Only
 self.K = K
def fit(self, X, y):
 # 2. Learn & Store State
 self.means_ = X.mean()
 return self
```

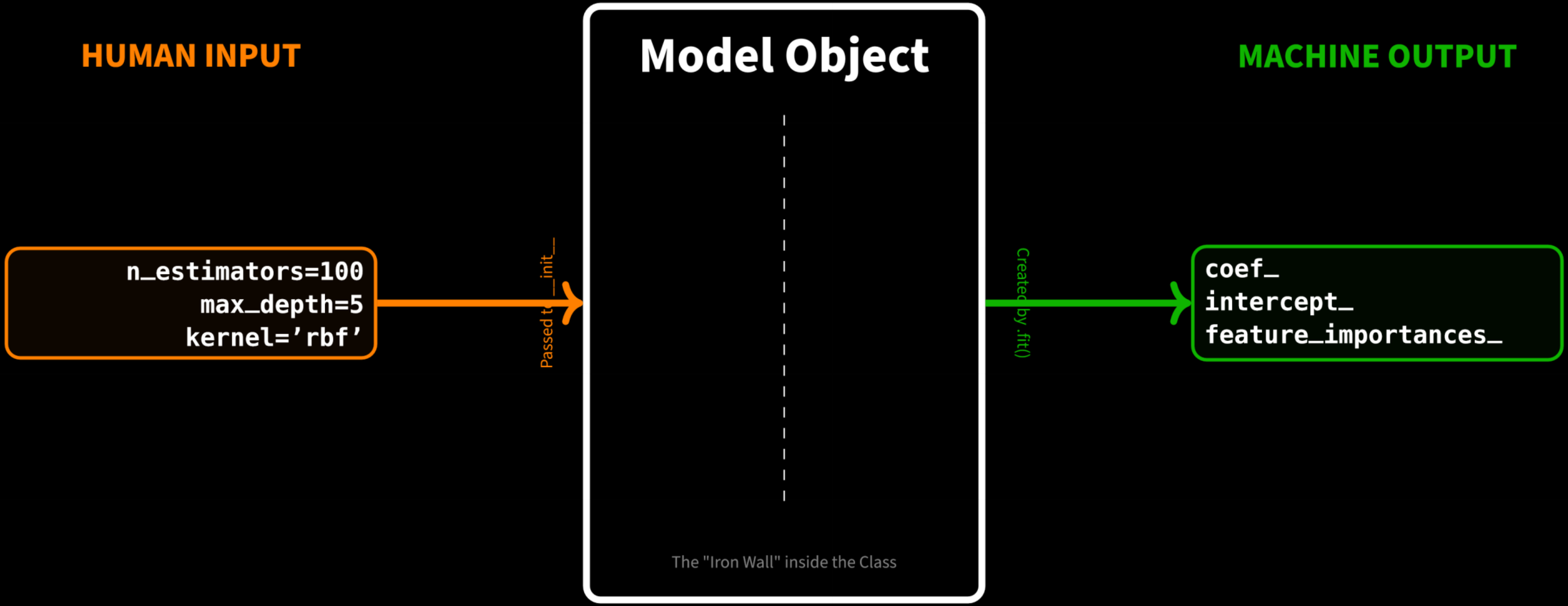
1.2 The Lifecycle: A State Machine

An Estimator transitions from **Stateless** (Empty) to **Stateful** (Fitted). Trying to jump the queue causes errors.



1.3 Anatomy of Parameters (The Difference)

Human Input vs **Machine Output**. If you confuse these, you cannot build custom models.

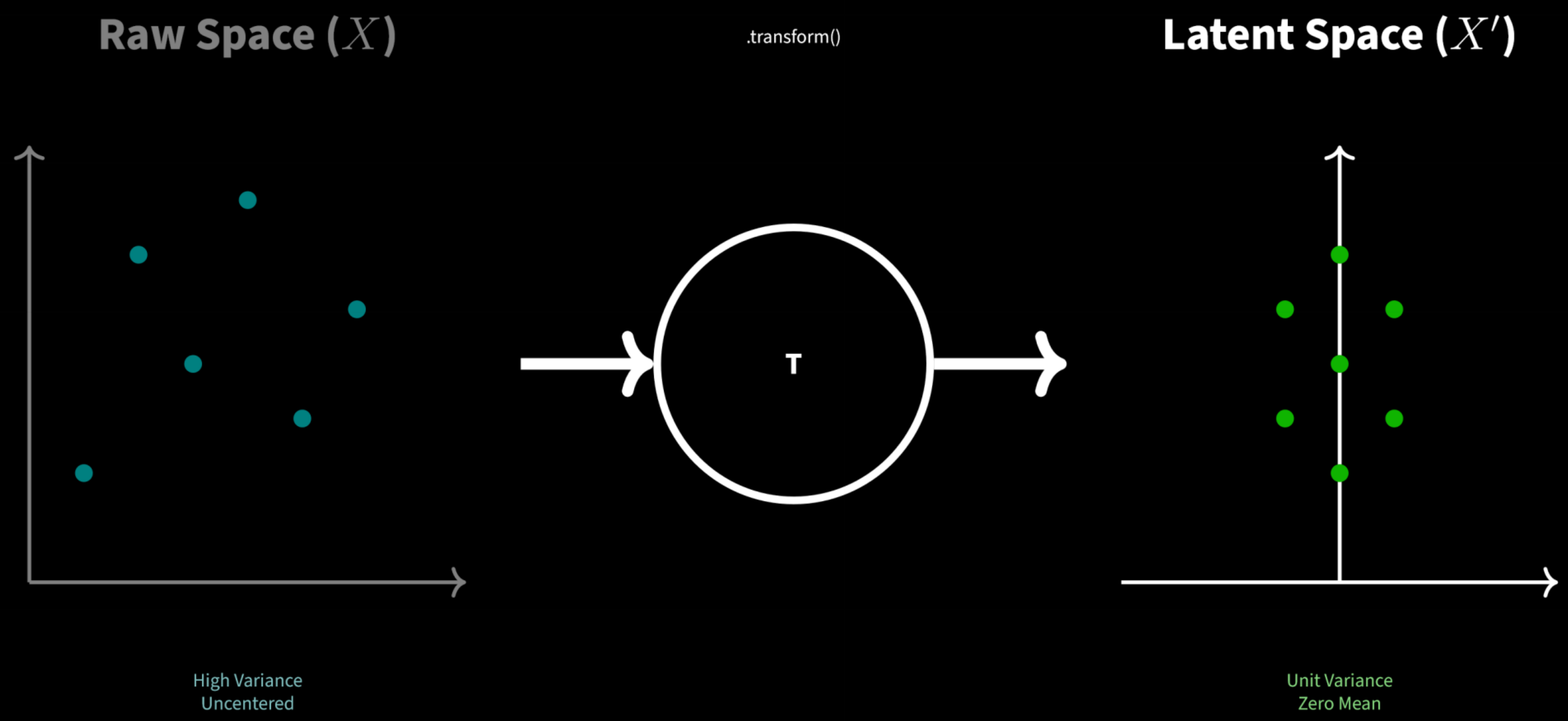


2. The Transformer

2.1 The Concept: Changing Representation

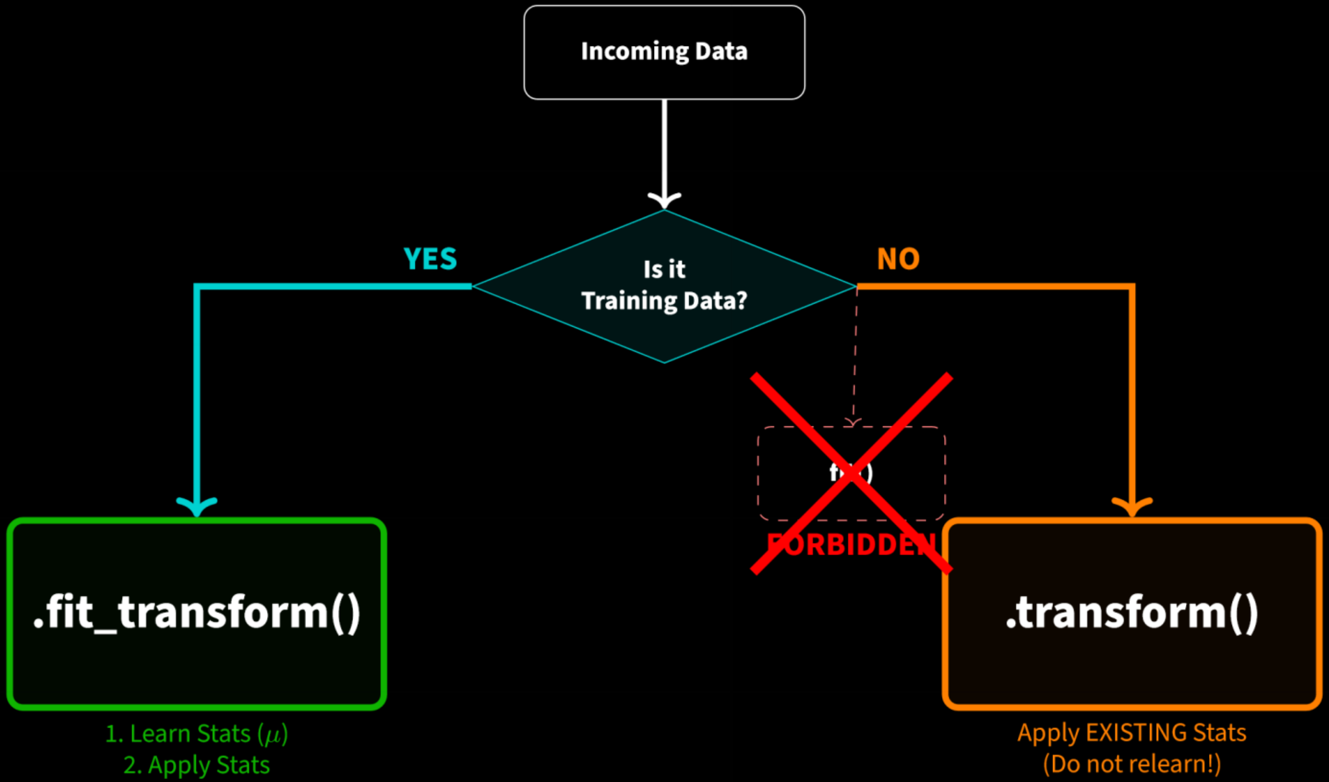
A Transformer does not predict labels. It changes the **geometry** of the feature space. It maps the input matrix X to a new, "cleaner" matrix X'.

- **StandardScaler:** Shifts and squishes distribution.
- **PCA:** Rotates and projects axes.


Raw Space (X)
.transform()
Latent Space (X')
T
High Variance
Uncentered
Unit Variance
Zero Mean

2.2 The Workflow: When to Fit vs Transform?

The Rule: *Teach* (Fit) on Training data. *Apply* (Transform) on everything else.



2.3 The Mechanics of Memory

The Transformer acts as a storage locker for dataset statistics. Once `fit()` is called, the "Brain" is frozen.

The "Brain"

```
self.mean_ = 5.2
self.scale_ = 1.1
self.n_samples_seen_ = 1000
```

Locked after Training

Retrieves Parameters

The Math Engine

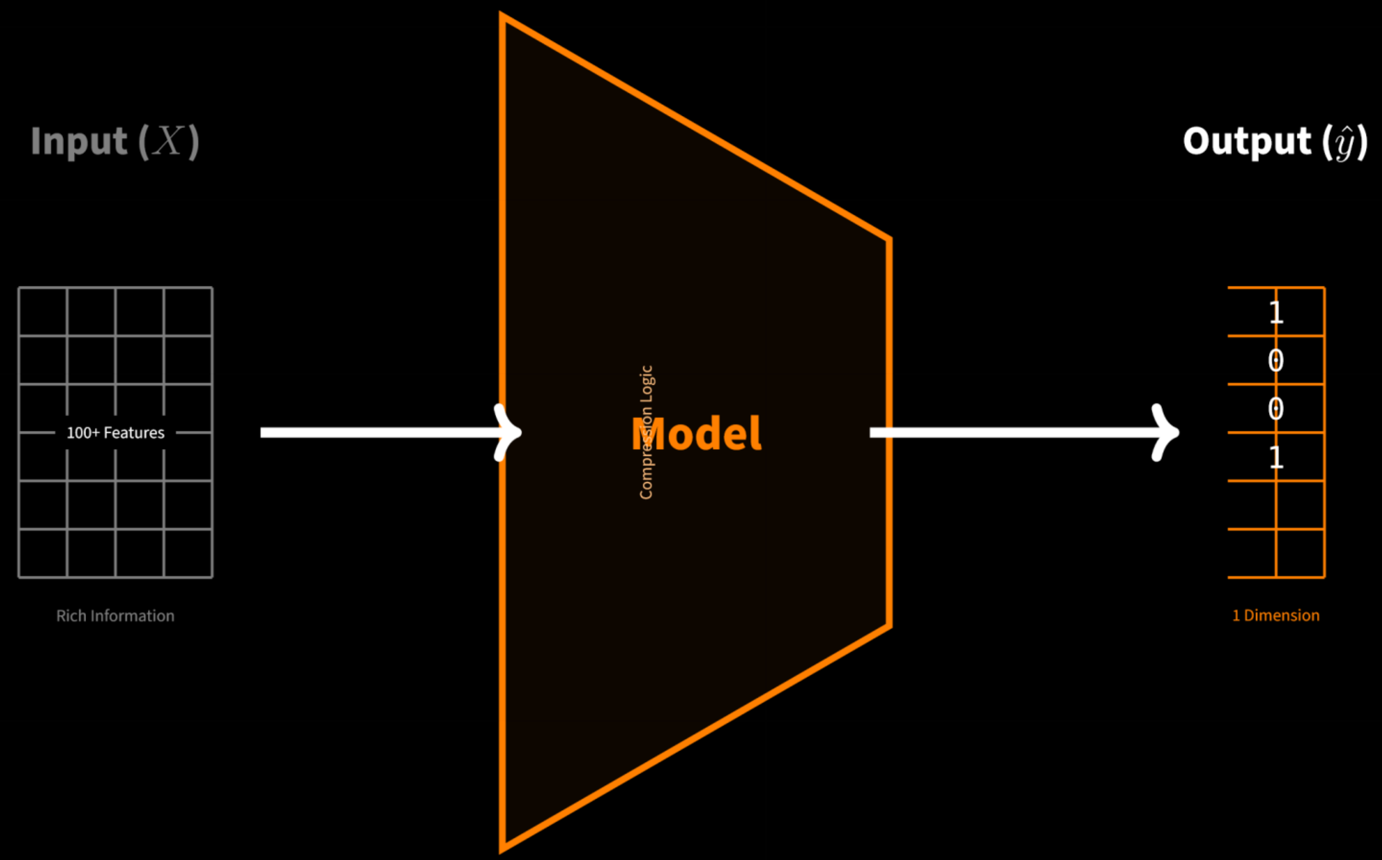
$$x_{new} = \frac{x - \text{mean\_}}{\text{scale\_}}$$

3. The Predictor

3.1 The Concept: Dimensionality Collapse

A Predictor is a funnel. It takes complex, high-dimensional reality (Features) and collapses it into a simple, low-dimensional decision (Labels).

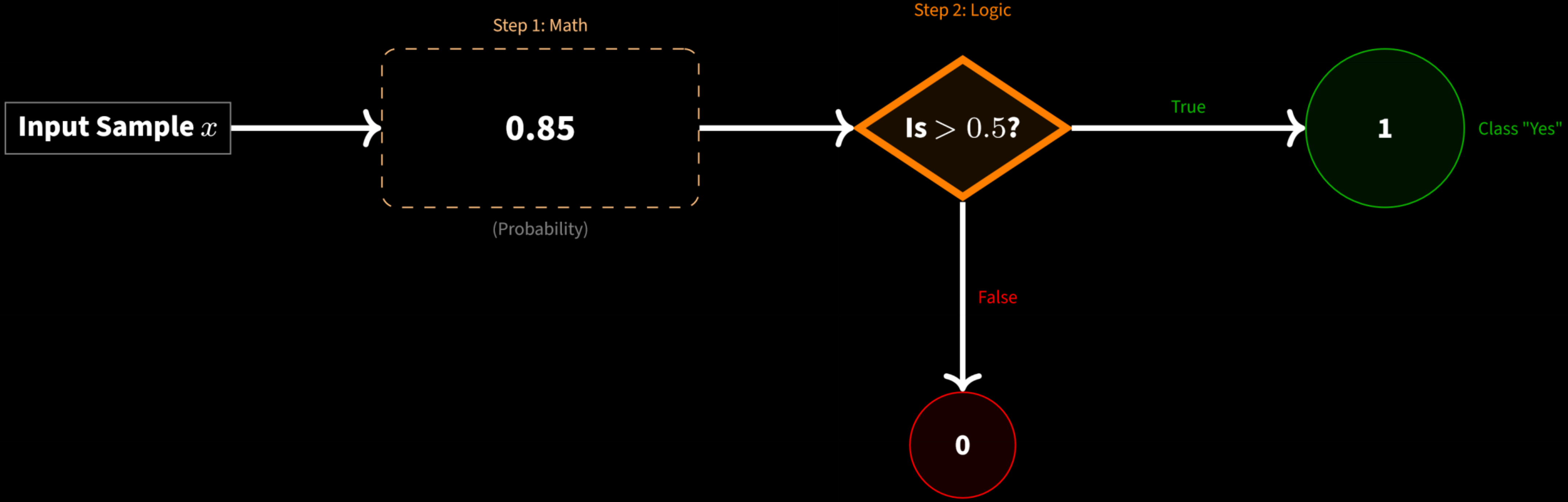
- **Regression:** Collapses to a continuous number (Quantity).
- **Classification:** Collapses to a discrete class (Category).



Input (X)
100+ Features
Rich Information
Model
Compression Logic
Output ($\hat{y}$)
1
0
0
1
1 Dimension

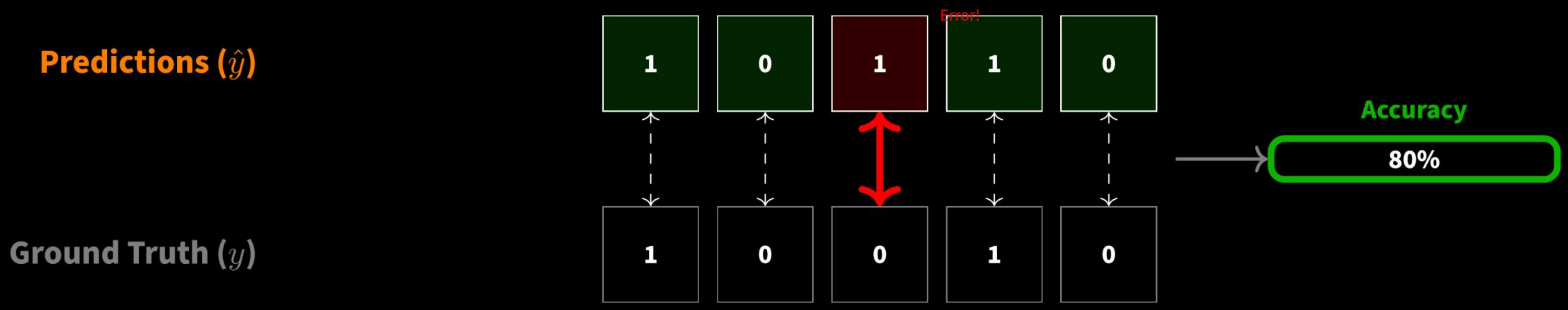
3.2 The Workflow: Inside .predict()

Most classifiers don't just "guess." They calculate a probability first. `.predict(X)` is actually a wrapper around `.predict_proba(X)` combined with a threshold check.



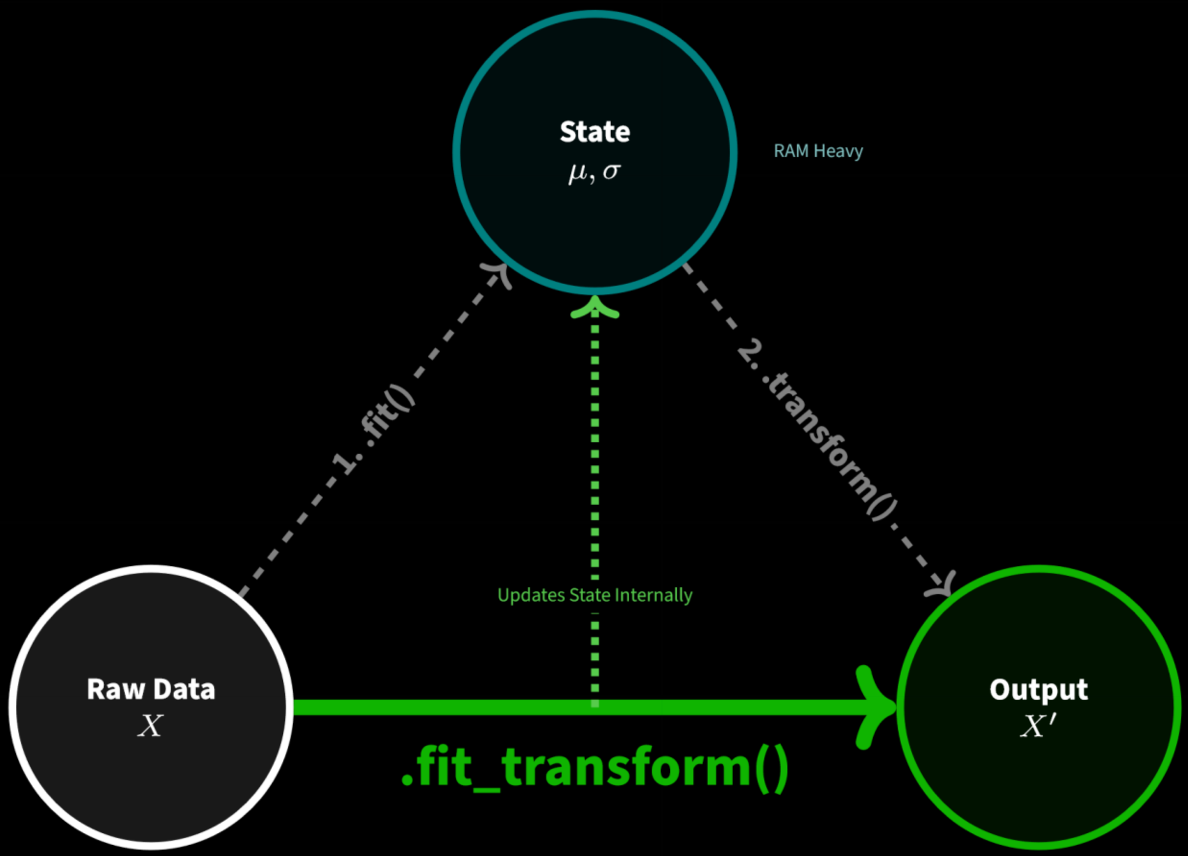
3.3 Validation: The Scoring Logic

How does `.score(X, y)` work? It generates a prediction vector internally and compares it element-by-element with the ground truth vector.



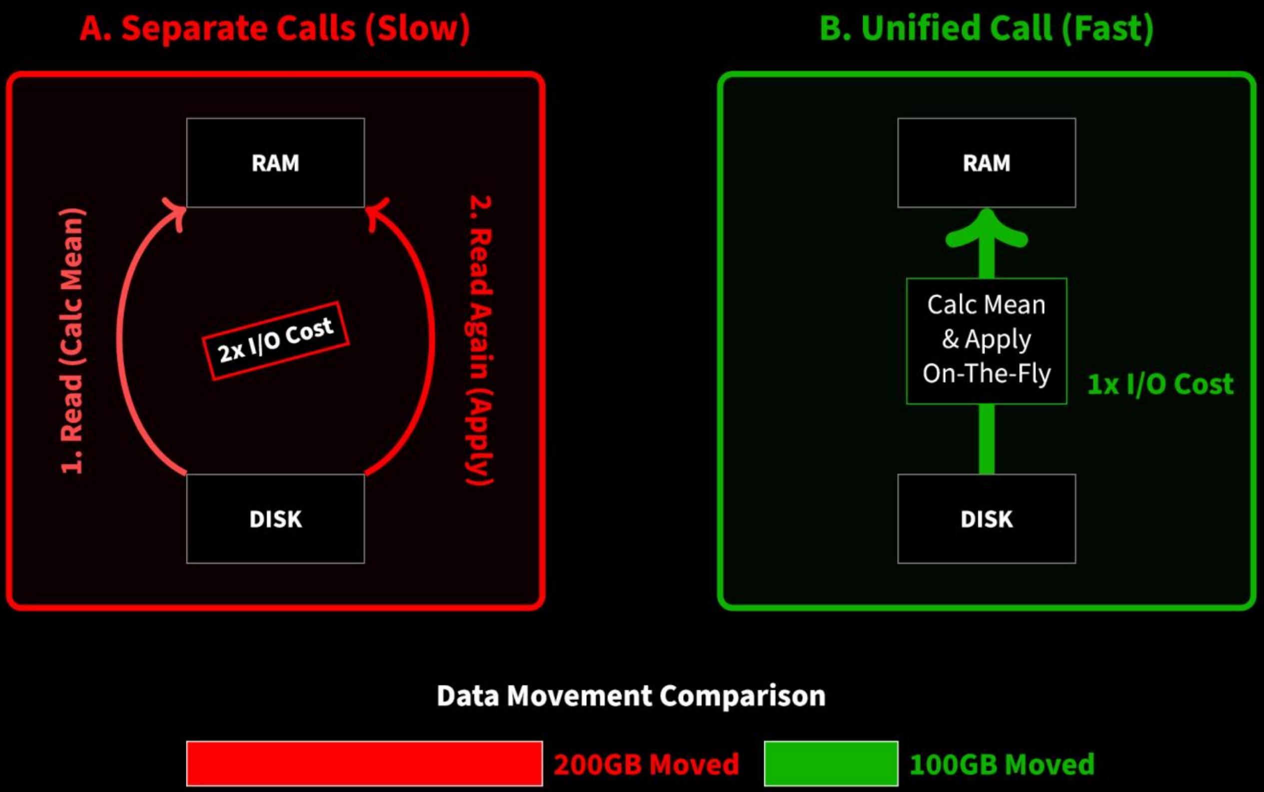
4.1 The Concept: The Efficiency Triangle

Geometry Rule: The straight line (Hypotenuse) is always faster than going around the sides.



4.2 The Mechanics: I/O Bottlenecks

The most expensive operation in Big Data is moving data from Disk to RAM. `fit_transform` prevents the "Double Loop" penalty.



4.3 The Pipeline Chain Reaction

When you use a `Pipeline`, Scikit-Learn automatically calls `fit_transform` on every step except the last one. This creates a "Bucket Brigade" where data is passed instantly from one worker to the next.